

Properties of Definite Integrals

Property-I

$$\int_a^b f(x) dx = \int_a^b f(t) dt = \int_a^b f(v) dv$$

Property-II

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

Property III

if $f(x)$ is continuous in $x \in [a, b]$

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx \quad \text{if } c \in (a, b)$$

Property IV

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

Proof:

$$\int_0^a f(a-x) dx$$

$\underbrace{\hspace{1cm}}_t$

$$a-x = t$$

$$-dx = dt$$

$$\int_a^0 f(t) dt = - \int_0^a f(t) dt = - \int_0^a f(x) dx$$

* Property V

* *

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

Property VI

$$\int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & \text{if } f(x) \text{ is even function} \\ 0 & \text{if } f(x) \text{ is Odd} \end{cases}$$

$$\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin x dx}{\sin x + \cos x}$$

$$f(x) = \frac{\sin x}{\sin x + \cos x}$$

Use: Property NO (5)

$$f\left(\frac{\pi}{2} - x\right) = \frac{\sin\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)}$$

$$I = \int_a^b f(x) dx$$

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin x dx}{\sin x + \cos x} \quad \text{--- (1)}$$

also: $I = \int_a^b f(a+b-x) dx$

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin\left(\frac{\pi}{2} - x\right) dx}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)}$$

$$I = \int_0^{\frac{\pi}{2}} \frac{\cos x dx}{\cos x + \sin x} \quad \text{--- (2)}$$

$$2I = \int_0^{\frac{\pi}{2}} 1 dx \quad (\because \text{Add eq (1) \& (2)})$$

$$2I = \frac{\pi}{2} - 0$$

$$I = \frac{\pi}{4}$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^7 x dx}{\sin^7 x + \cos^7 x}$$

$$\text{let } I = \int_0^{\frac{\pi}{2}} \frac{\sin^7 x dx}{\sin^7 x + \cos^7 x}$$

Use Property (5) $f\left(\frac{\pi}{2} - x\right)$

$$\int_{3/8}^1 [-8x^2 + 6x - 1] dx + \int_{3/8}^1 [-8x^2 + 6x - 1] dx$$

$$x=0 \Rightarrow -8x^2 + 6x - 1 = -1$$

$$x = \frac{3}{8} \Rightarrow -8x^2 + 6x - 1 = \frac{1}{8}$$

$$8x^2 + 6x - 1 = 1 \Rightarrow \boxed{x = \frac{3 \pm \sqrt{17}}{8}}$$

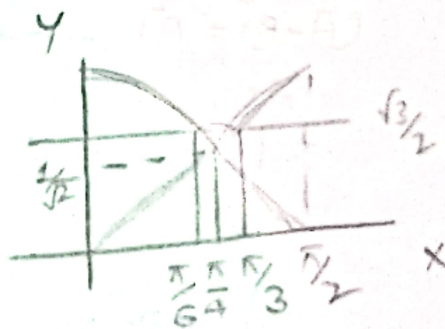
$$= \frac{3 \pm \sqrt{17}}{8} \int_{-1}^{\frac{3 \pm \sqrt{17}}{8}} [-8x^2 + 6x - 1] dx + \int_{\frac{3 \pm \sqrt{17}}{8}}^1 [-8x^2 + 6x - 1] dx + \int_{\frac{3 \pm \sqrt{17}}{8}}^1 [-8x^2 + 6x - 1] dx$$

MINIMUM OF $f_1(x)$, $f_2(x)$ OR
MAXIMUM OF $f_1(x)$, $f_2(x)$.

Q. Let $f(x) = \max \left\{ \sin x, \frac{\sqrt{3}}{2}, \cos x \right\}$ then

$$\int_0^{\frac{\pi}{2}} f(x) dx = ?$$

$$f(x) = \begin{cases} \cos x & x \in (0, \frac{\pi}{6}) \\ \frac{\sqrt{3}}{2} & x \in (\frac{\pi}{6}, \frac{\pi}{3}) \\ \sin x & x \in (\frac{\pi}{3}, \frac{\pi}{2}) \end{cases}$$



$$\int \cos x + \int \frac{\sqrt{3}}{2} dx + \int \sin x dx$$

$$= \sin x + \frac{\sqrt{3}}{2} x + (-\cos x)$$

$$= \frac{1}{2} + \frac{\sqrt{3}}{2} \left(\frac{\pi}{6} \right) - \left(0 - \frac{1}{2} \right)$$

$$\text{Hence } 1 + \frac{\sqrt{3}\pi}{12}$$

$$= \frac{1}{2} + \frac{\sqrt{3}\pi}{12} \Rightarrow \frac{6 + \sqrt{3}\pi}{12}$$

$$Q. \int_0^{10\pi} [\tan^{-1}x] dx$$

$$\tan^{-1}x = 1$$

$$x = \tan 1$$

$$\int_0^{\tan 1} [\tan^{-1}x] dx + \int_{\tan 1}^{10\pi} [\tan^{-1}x] dx$$

$$= 10\pi - \tan 1$$

$$\tan 45 = 1$$

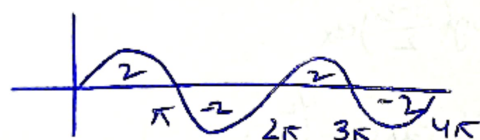
$$\tan 60 = \frac{\sqrt{3}}{3} = \sqrt{3} = 1.732$$

$$\tan 90 = \frac{\pi}{2} = \infty$$

Property-8:

If $f(x)$ is continuous in (a, b) and $\int_a^b f(x) dx = 0$ then $f(x)$ will have at least one root b/w (a, b) .

Ex: $\sin(x) = f(x)$ for $x \in (0, 4\pi)$ Prove that $\sin x$ will have at least one root.



$$\int_0^{4\pi} (\sin x) dx = 0$$

Q. If $f(x)$ is continuous and

$$\int_0^2 f(ax^2+bx+c) \cdot (1+\cos^8 x) dx = \int_0^1 f(ax^2+bx+c) \cdot (1+\cos^8 x) dx$$

Prove that $f(ax^2+bx+c)$ has at least one root in $(1/2)$

$$\int_0^2 f(ax^2+bx+c) \cdot (1+\cos^8 x) dx + \int_0^2 f(ax^2+bx+c) \cdot (1+\cos^8 x) dx = \int_0^1 f(ax^2+bx+c) \cdot (1+\cos^8 x) dx$$

$$\int_0^2 f(ax^2+bx+c) \cdot (1+\cos^8 x) dx = 0$$

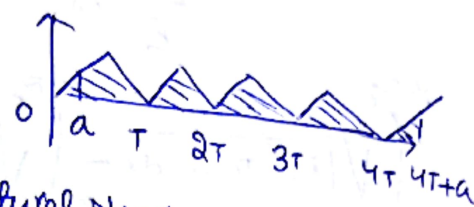
Use Property NO. 8.

Property - (9)

If $f(x)$ is periodic function with Periodic T

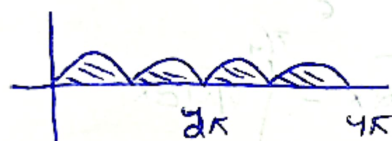
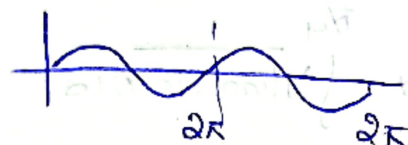
$$\int_0^T f(x) dx = \int_0^T f(x) dx \quad **$$

$$\int_a^{a+nT} f(x) dx = n \int_a^T f(x) dx$$



n is Natural Numbers !

$$Q: \int_0^{4\pi} |\sin x| dx$$



$$= 4 \int_0^{\pi} |\sin x| dx$$

$$= 4 \int_0^{\pi} \sin x dx = 4 [-\cos x]_0^{\pi}$$

$$= 4 [1 - (-\cos 0)]$$

$$= 4 [2]$$

$$= 8$$

$$Q: \int_0^{5\pi} |\tan^{-1}(\tan x)| dx = 5 \int_0^{\pi} |\tan^{-1}(\tan x)| dx$$

$$= 10 \int_0^{\pi/2} |\tan^{-1}(\tan x)| dx = 10 \int_0^{\pi/2} x dx$$

$$= \frac{5\pi^2}{4}$$

Property (10):

If $f(x)$ is Strictly Monotonic in (a, b) then

$$\int_a^b f(x) dx + \int_{f(a)}^{f(b)} f^{-1}(x) dx = bf(b) - af(a)$$

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$$\int_1^2 \log_2(x^3+1) dx + \int_1^{\log_2 9} (2^x-1)^{1/3} dx = ?$$

Ans:

$$\log_2(x^3+1) = y$$

$$x^3+1 = 2^y$$

$$x = (2^y-1)^{1/3}$$

$$= 2[2^3-1] - 1[1^3-1]$$

$$= 2[7] - 0$$

$$= 14$$

$$f(x) = \log_2(x^3+1) \Rightarrow 2 \log_2 9 - \log_2 2$$

$$\Rightarrow \log_2 \frac{81}{2}$$

$$\frac{40.5}{2}$$

• Find nearest Perfect Square

$$32 < \frac{81}{2} < 64$$

$$\log_2 32 < \log_2 \frac{81}{2} < \log_2 64$$

$$5 < \log_2 \frac{81}{2} < 6 \text{ Hence } \left\lfloor \log_2 \frac{81}{2} \right\rfloor \text{ integral value is } 5$$

So

$$\int (\cos x + \sin x) dx$$

0.

?

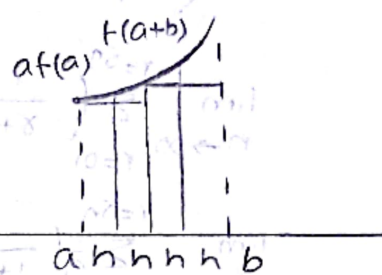
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Limit of a Sum as definite integral:

$$\int_a^b f(x) dx = \text{Area Under the Curve}$$

$$A_1 = hf(a) + h(f(a+h)) < \int_a^b f(x) dx$$

$$A_2 = hf(a) + h(f(a+h)) + h(f(a+2h)) + h(f(a+3h)) + \dots < \int_a^b f(x) dx$$



Divide $b-a$ into n equal lengths:

let it be h $h = \frac{b-a}{n}$

Sum of Areas of rectangles

$$= hf(a) + h(f(a+h)) + h(f(a+2h)) + \dots + h(f(a+(n-1)h))$$

$$= h [f(a) + f(a+h) + f(a+2h) + \dots + f(a+(n-1)h)]$$

$$= h \sum_{r=0}^{n-1} f(a+rh)$$

$$S = \frac{b-a}{n} \sum_{r=0}^{n-1} f(a + (\frac{b-a}{n})r)$$

$$= \lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{r=0}^{n-1} f(a + r(\frac{b-a}{n})) = \int_a^b f(x) dx$$

Note:

$$\lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{r=0}^{n-1} f(a + r(\frac{b-a}{n}))$$

$$a + r(\frac{b-a}{n}) = x$$

$$\frac{b-a}{n} = dx, \quad \sum \rightarrow \int$$

$$\Rightarrow \int_a^b f(x) dx$$

$\Rightarrow$ RHS ...

Newton-leibnitz rule :

$$\frac{d}{dx} \int_{h(x)}^{g(x)} f(t) dt = f(g(x))g'(x) - f(h(x)) \cdot h'(x)$$

Example :

$$F(x) = \int_{x^2}^{x^3} \cos t^2 dt, \text{ find } F'(x)$$

$$\Rightarrow \cos x^6 \cdot 3x^2 - \cos x^4 \cdot 2x$$

$$\Rightarrow 3x^2 \cos x^6 - 2x \cos x^4$$

$$1. \lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sin t^2 dt}{x^6}$$

$$= \frac{\sin x^4 \cdot 2x - \sin 0 \cdot 0}{6x^5}$$

$$= \frac{2x \sin x^4}{6x^5}$$

$$= \frac{x \sin x^4}{3x^4}$$

$$= \frac{\sin x^4}{x^3} \left(\frac{1}{3} \right) = \left(\frac{1}{3} \right)$$

DPP-3

10 11 12 x

$$= \lim_{x \rightarrow 1} \frac{\int \ln(3t - 3t^2 + t^3) dt}{(1-x)^3}$$

$$= \lim_{x \rightarrow 1} \frac{\ln(3x - 3x^2 + x^3) \cdot (1) - \ln(3 - 3 + 1) \cdot 0}{-3(1-x)^2 \cdot (-1/x)}$$

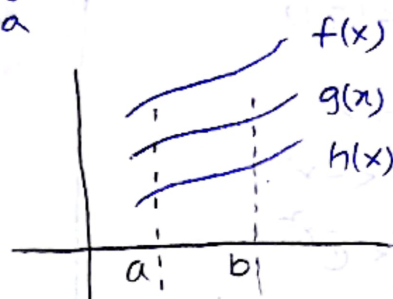
$$= \frac{1}{3} \lim_{x \rightarrow 1} \frac{\ln(3x - 3x^2 + x^3)}{(1-x)^2}$$

$$= \frac{1}{3} \lim_{x \rightarrow 1} \left(\frac{3x - 3x^2 + x^3}{2(1-x)^2} \right) \Rightarrow \frac{1}{3} \lim_{x \rightarrow 1} \left(\frac{3x - 3x^2 + x^3}{2(1-x)^2} \right)$$

Property-11:

if $f(x)$, $g(x)$, $h(x)$ are 3 continuous functions
 $x \in (a, b)$ Such that $f(x) \geq g(x) \geq h(x)$ then

$$\int_a^b f(x) dx \geq \int_a^b g(x) dx \geq \int_a^b h(x) dx$$



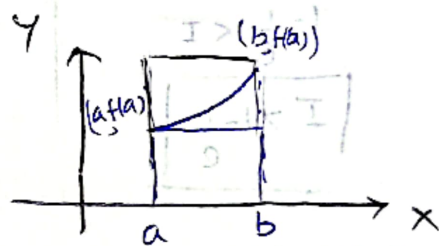
Property-12:

let $f(x)$ is strictly increasing in $x \in (a, b)$

$$(b-a)f(a) < \int_a^b f(x) dx < (b-a)f(b)$$

$$\Rightarrow \int_a^b f(x) dx > (b-a)f(a)$$

$$\Rightarrow \int_a^b f(x) dx < (b-a)f(b)$$



Q. Prove that $e-1 > \int_0^1 e^{x^2} dx > 1$

Proof:

$$b-a(f(b)) > \int_a^b f(x) dx > \frac{1}{2}(b-a)f(a)$$

$$(1-0)f(1) > \int_0^1 e^{x^2} dx > (1-0)f(0)$$

$$e > \int_0^1 e^{x^2} dx > 1$$

$$x \in (0, 1)$$

$$x > x^2$$

$$e^x > e^{x^2}$$

$$\int_0^1 e^x dx > \int_0^1 e^{x^2} dx$$

$$(e^x)'_0 > 1$$

$$(e-1) > 1$$

$$\int_K^{K+1} \frac{dx}{x(x+1)} < \int_K^{K+1} \frac{dx}{x(K+1)}$$

$$(K+1) \int_K^{K+1} \frac{dx}{x(x+1)} < \int_K^{K+1} \frac{dx}{(K+1)x}$$

$$\sum_{K=1}^{98} (K+1) \int_K^{K+1} \frac{dx}{x(x+1)} < \sum_{K=1}^{98} \int_K^{K+1} \frac{dx}{x}$$

$$I < \sum_{K=1}^{98} (\log x)_K^{K+1}$$

$$I < \sum_{K=1}^{98} (\log K+1 - \log K)$$

$$I < [\log 2 - \log 1 + \log 3 - \log 2 + \log 4 - \log 3 + \dots + (\log 99 - \log 98)]$$

$$I < \log 99$$

Walli's Formula:

$$\begin{aligned} \text{ii. } \int_0^{\pi/2} \sin^n x dx &= \int_0^{\pi/2} (\cos x)^n dx \\ &= \left(\frac{(n-1)(n-3) \dots}{n(n-2) \dots} \right) \frac{\pi}{2} \quad \text{if } n \text{ is Even} \end{aligned}$$

$$= \left(\frac{(n-1)(n-3) \dots}{n(n-2) \dots} \right) \quad \text{if } n \text{ is Odd}$$

$$I_n = \int_0^{\pi/2} \sin^n x dx$$

$$I_n = \int_0^{\pi/2} \sin^{n-1} x \cdot \sin x dx \quad (\text{By Parts})$$

$$\begin{aligned} \text{1Q. } \int_0^{\pi/2} \sin^7 x dx &= \frac{(6)(4)(2)}{(7)(5)(3)(1)} = \frac{16}{35} \\ \text{2Q. } \int_0^{\pi/2} \sin^8 x dx &= \frac{35\pi}{256} \end{aligned}$$